

Student Learning Outcomes:

1. **Describe the Structure of the CCEI 2.0.**
2. **Apply Operational Definitions to Competency Assessment**
3. **Differentiate Scoring Approaches and Assessment Types**
4. **Evaluate the Role of Feedback in Competency Development**

Welcome everyone! I'm excited to guide you through this module on the Creighton Competency Evaluation Instrument 2.0. I'll offer a **comprehensive overview of how this tool is structured and utilized to assess competency in nursing education.**

You have explored the conceptual framework that guided the development of the CCEI 2.0, which ensures that competency assessment is rigorous and aligned with best practices. Now, this module will shift focus to the structure of the instrument.

Let's take a closer look at the structure. This **instrument is organized into four major categories for a total of 25 items.** For this instrument, the developers aimed to create a comprehensive tool for evaluating student performance, ensuring it goes beyond narrow assessments like clinical judgment alone or skill acquisition. Instead, the CCEI 2.0 intentionally covers broad areas of nursing practice, all rooted in the conceptual frameworks presented in the previous module.

However, before using an **assessment tool** like the **C-CEI 2.0** or other competency-based evaluation framework, it is **essential to understand the operational definitions of each item.** This ensures that assessments are **consistent, objective, valid, and aligned with competency expectations**, ultimately supporting **student learning and professional readiness** in nursing practice. I will include the definitions of the categories, subcategories, and items as I talk through each of these areas.

I will start with the 4 major categories that frame this instrument: Clinical Judgment, Communication, Quality and Safety, and Professionalism)

(?show definition of clinical judgment while presenting this script – definition below at end of this section in table)

Clinical judgment is the cornerstone of safe, effective, and evidence-based patient care. The National Council of State Boards of Nursing defines clinical judgment as “the skill of recognizing cues regarding a clinical situation, generating and weighing hypotheses, taking action, and evaluating outcomes for the purpose of arriving at a satisfactory clinical

outcome. Clinical judgment is the observed outcome of two unobserved underlying mental processes, critical thinking and decision making.”

In our healthcare setting, **both prelicensure and advanced practice students must be able to assess, analyze, and respond to dynamic patient conditions.** With permission from the National Council of State Board of Nursing to use their clinical judgment measurement model as part of our framework, we have developed an essential tool for faculty to evaluate this critical concept in both prelicensure and advance practice students. Including clinical judgment in our evaluation instrument reinforces its central role in practice readiness, aligns with national nursing education standards, and provides faculty with a structured approach to assess students’ competency development across a program.

Category	Definition
Clinical Judgment	“The skill of recognizing cues regarding a clinical situation, generating and weighing hypotheses, taking action, and evaluating outcomes for the purpose of arriving at a satisfactory clinical outcome. Clinical judgment is the observed outcome of two unobserved underlying mental processes, critical thinking and decision making” (NCSBN, 2018)

(?show definition of communication while presenting this script – definition below in table)

Effective communication is fundamental to safe, high-quality, and patient-centered care. The American Association of Colleges of Nursing defines communication as “...an exchange of information, thoughts, and feelings through a variety of mechanisms. The definition encompasses the various ways that people interact with each other, including verbal, written, behavioral, body language, touch, and emotion. Communication also incorporates intentionality, mutuality, partnerships, trust, and presence. Effective communication....is required for high quality, individualized health care...and can be delivered through technological modalities...”

Given the complexity of healthcare, prelicensure and nurse practitioner students must develop strong communication skills to effectively collaborate with patients, families, interdisciplinary teams, and other stakeholders. **Including communication as a category in the CCEI 2.0 provides faculty with a structured means to assess student’s ability to convey critical**

information, demonstrate empathy, build therapeutic relationships, document effectively, and adapt their communication approach based on the situation and available technology.

This category reinforces the essential role of communication in clinical practice, aligns with national nursing education standards, and ensures faculty can evaluate students' competency in this foundational skill, which is central to the nursing profession.

Category	Definition
Communication	"...an exchange of information, thoughts, and feelings through a variety of mechanisms. The definition encompasses the various ways that people interact with each other, including verbal, written, behavioral, body language, touch, and emotion. Communication also incorporates intentionality, mutuality, partnerships, trust, and presence. Effective communication....is required for high quality, individualized health care...and can be delivered through technological modalities..." (AACN, 2021)

(?show definition of Quality and Safety while presenting this script – below in table)

Ensuring quality and safety in patient care is a fundamental responsibility of every nurse. The Quality and Safety Education for Nurses (QSEN) defines quality and safety as: The individual acts to provide safe care by minimizing risk of harm to clients and providers through both system effectiveness and individual performance. Includes using data to evaluate the outcomes of the care processes and uses improvement strategies to make changes for continuously safe and quality care.

By including Quality and Safety as a category in the CCEI 2.0, we provide faculty with a structured way to assess students' ability to integrate informatics to support clinical decision-making, adhere to patient safety principles, and advocate for clients within complex healthcare systems. This category reinforces the importance of continuously improving care processes through data-driven strategies while ensuring that students develop the skills necessary to minimize risks and promote optimal patient outcomes. Aligning with national nursing education standards, this evaluation framework supports the preparation of nurse practitioners and prelicensure students who are committed to delivering high-quality, safe, and evidence-based care.

Category	Definition
Quality and Safety	The individual acts to provide safe care by minimizing risk of harm to clients and providers through both system effectiveness and individual performance. Includes using data to evaluate the outcomes of the care processes and uses improvement strategies to make changes for continuously safe and quality care (QSEN, 2020).

(?show definition of Professionalism while presenting this script – definition below in table)

Professionalism is a fundamental expectation in nursing practice, shaping the way nurses interact with patients, co-workers, and the broader healthcare system. The definition of professionalism as defined by the American Association of College of Nursing states: “Professionalism encompasses the development of a nursing identity embracing the values of integrity, altruism, inclusivity, compassion, courage, humility, advocacy, caring, autonomy, humanity and social justice. Professionalism necessitates the development of emotional intelligence to promote social good, engage in social justice, and demonstrate ethical comportment, moral courage and assertiveness in decision-making and actions”

Including Professionalism as a category in the CCEI 2.0 ensures that faculty have a structured way to assess students’ ability to adhere to legal and ethical standards, demonstrate accountability, and uphold the values central to the nursing profession. This category reinforces the importance of emotional intelligence, ethical decision-making, and advocacy in clinical practice while aligning with national nursing education standards.

Professionalism	“Professionalism encompasses the development of a nursing identity embracing the values of integrity, altruism, inclusivity, compassion, courage, humility, advocacy, caring, autonomy, humanity and social justice. Professionalism necessitates the development of emotional intelligence to promote social good, engage in social justice, and demonstrate ethical comportment, moral courage and assertiveness in decision-making and actions” (AACN, 2020).
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By including these four categories, the developers of the CCEI 2.0 feel that we have developed an instrument that reflects the conceptual definition of clinical competency in nursing.

(?Include visual of the NCSBN clinical judgment measurement model ?? - then visual of our definitions during the script.)

Now, I will discuss the six subcategories that further divide out the concepts within the category of Clinical Judgment. As mentioned earlier, Clinical judgment is essential for safe, effective, and evidence-based nursing practice. To ensure students develop this critical competency, with permission from the National Council of State Boards of Nursing's, the CCEI 2.0 incorporates their Clinical Judgment Measurement Model (CJMM) as a framework for assessing clinical judgment. **This model provides a structured approach to evaluating clinical decision-making by breaking it down into six key subcategories and provides a valuable structure for educators to provide student feedback and to implement targeted remediation.**

Within the Clinical Judgment category, there are six subcategories: Recognize Cues, Analyze Cues, Prioritize Hypotheses, Generate Solutions, Take Actions, and Evaluate Outcomes.

(should I read this brief overview of the 6 subcategories while they read the actual definitions on the screen?? - or read the definitions? - full definitions below in table)

- **Recognize cues** – Identifying relevant clinical data and patient information.
- **Analyze cues** – Interpreting the significance of the gathered information.
- **Prioritize hypotheses** – Determining the most likely explanations for a patient's condition.
- **Generate solutions** – Developing appropriate interventions based on clinical reasoning.
- **Take actions** – Implementing interventions to address patient needs.
- **Evaluate outcomes** – Assessing the effectiveness of interventions and adjusting care as needed.

By using this model as part of the CCEI 2.0, we ensure a comprehensive and standardized method for evaluating students' ability to apply clinical judgment in a variety of scenarios. **This framework not only aligns with national education standards but also strengthens students' readiness for practice by emphasizing a systematic approach to decision-making.** While the CJMM was originally developed to assess prelicensure nursing students, its structured framework for evaluating clinical judgment is **equally**

valuable in assessing nurse practitioner students. Advanced practice nurses must demonstrate a high level of critical thinking and decision-making skills to provide safe and effective care. **The CJMM's six categories – Recognize cues, analyze cues, prioritize hypotheses, generate solutions, take actions, and evaluate outcomes all align with the complex clinical reasoning processes required for advanced practice.** Using the CJMM, faculty can evaluate **nurse practitioner students' ability to synthesize complex patient data, develop differential diagnoses, formulate evidence-based treatment plans and assess patient outcomes.** This model ensures consistency in evaluating clinical judgment **across all levels of nursing education.**

CCEI 2.0 Definitions of Subcategories of Clinical Judgment

Subcategory	Definition
Recognize Cues	The learner recognizes pertinent cues from clients, families, or populations and can identify variances when they occur. Cues may come from subjective or objective data, including the physical environment and social determinants of health (personal, social, economic, and environmental factors).
Analyze Cues	The learner analyzes cues from the subjective and objective data. As the learner interprets the data, they identify potential cause(s) of clinical problems. This includes a synthesis of the assessment data within the context of the clinical setting
Prioritize Hypotheses	The learner evaluates and ranks identified concerns. This prioritization can take many forms, such as urgency of concern, identifying cause of current symptoms, and/or determining what is the greatest risk factor contributing to current condition.
Generate Solutions	During this phase, the learner is creating solutions to address the identified concerns. Generation of solutions is informed by continuous recognition & analysis of new information as it becomes available, including input from the client. This can and should include client input into the plan of care, as appropriate.
Take Actions	To act in a particular way to ensure a specific result. This includes the implementation of interventions/consultations to ensure a specific outcome for the clinical problem. These interventions should be specific to the patient and their presentations for the identified clinical problem(s) (Betts et al., 2019). Ensure that all interventions utilized are documented appropriately.
Evaluate Outcomes	Measuring the effectiveness of interventions used to treat/improve the identified clinical problem(s). Ensure that the observed outcomes are consistent with the expected outcomes of the interventions used. If the observed outcomes are not

	consistent with the expected outcomes, then new interventions should be considered (Betts et al., 2019). Ensure that outcomes are appropriately documented.
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Now that I have covered the operational definitions for the subcategories in clinical judgment, I will shift to cover the **operational definitions for all of the 25 items**.

The operational definitions at the level of the items are where the distinction can be made between assessing prelicensure and advanced practice students. The highlighted cells on the chart indicate where the definitions differ between Prelicensure and Advanced Practice nursing students – and both definitions are presented side-by-side with the definition for prelicensure students on the left and nurse practitioner students on the right.

??best way to present this information??

Item No	Item	Pre-Licensure	Advanced Practice
Category: Clinical Judgment			
Subcategory: Recognize Cues			
1	Obtains pertinent subjective data	Identifies pertinent history information from client. Considers all aspects of the health history. Conducts the health history in organized, logical manner. Asks probing questions, reviews medical record. Uses appropriate screening tools.	
2	Obtains pertinent objective data	Identifies pertinent physical exam data. Uses techniques of inspection, palpation, auscultation, and percussion appropriately. Obtains vital signs, lab data, physical examination data in a systematic and organized manner.	
3	Assesses the environment	Assesses the environment, dependent upon scenario and emphasis of the scenario. This can be the physical environment of a hospital room, a clinic room, a scene, an ambulance, a home environment, a community environment, a structural or unit environment.	
4	Assesses Factors Affecting Health	Assesses for any factors influencing health, especially spiritual, cultural, and social determinants of health. Examples could include: Assess housing instability, food insecurity, transportation problems, utility help needs, interpersonal safety. Uses appropriate screening tools.	
5	Recognizes variances/pertinent data	Differentiates between normal and abnormal presenting symptoms, observations, or facts.	
Subcategory: Analyze Cues			
6	Interprets subjective data	Identifies meaning of subjective finding (patient report of a symptom, piece of family history data from electronic health	

		record) and can cluster finding with other like findings (or separate from other findings).	
7	Interprets objective data	Identifies the meaning of objective finding (lung sound, wound appearance, lab value) and can cluster finding with other like findings (or separate from other findings).	
8	Identifies cause(s) of problem	Formulates hypotheses and appraisals to identify the cause of the identified problem.	Explores potential causes of problem(s); formulates list of appropriate differential diagnosis based on clinical presentation.
Subcategory: Prioritize Hypotheses			
9	Identifies expected outcomes	Identifies outcomes related to possible differential diagnoses. This can be informed by asking “What is the greatest risk?”, or “What is urgent?”, or “What outcome takes priority?” This could require a ‘pause’ in scenario where student generates a list of priority outcomes.	
10	Develops/Prioritizes interventions	Develops a list of potential interventions to assist client. Prioritizes the potential interventions.	Develops and Prioritizes differential diagnoses appropriately.
Subcategory: Generate Solutions			
11	Generates plan	Creates plan of care strategies to address priority outcomes and/or problems. This can include interventions and education, etc.	Rules out differentials through appropriate diagnostic testing or other strategy. Choose final diagnosis if appropriate. Creates management plan to address priority outcomes and/or problems. This can include interventions and education, etc.
12	Reviews/reassesses pertinent data	Reassesses patient data as needed to inform and/or reprioritize solutions/interventions.	Reassesses differential diagnosis list periodically as needed. Uses patient data as needed to inform and/or reprioritize differentials or change diagnosis.
Subcategory: Take Actions			
13	Manages equipment/devices involved in care.	Manages all equipment that is being used in care of client appropriately and safely (for example: otoscopes, ophthalmoscopes, IV pumps, lifts, beds, etc.)	
14	Implements evidence-based interventions	Performs interventions correctly and safely, provides rationale (evidence) for interventions (includes patient education).	
Subcategory: Evaluate Outcomes			

15	<i>Evaluates interventions in relation to expected outcomes</i>	<i>Evaluates if interventions were effective in addressing outcomes and/or problems. May require a 'pause' in scenario for learner to discuss/list what assessment findings are leading them to conclusion that intervention was either effective/ineffective.</i>
16	<i>Modifies expected outcomes, assessments, and interventions</i>	<i>Modifies outcomes or assessments and interventions based upon if outcomes were achieved. May require a 'pause' in scenario for learner to discuss/list what modifications would be necessary.</i>
17	<i>Demonstrates reflective practice</i>	<i>Provides written or verbal reflection on personal strengths and opportunities demonstrated in performance in the experience. Identifies near misses, mistakes made, and how to avoid in the future. Articulates plan for improvement and what was learned through the experience. Demonstrates insight into own biases, assumptions, actions.</i>
Category: Communication		
18	<i>Communicates professionally.</i>	<i>Engages professionally in all interactions. Stakeholders can include a variety of audiences: client, family/support, leaders, community agencies, policymakers, sources of funding, nurses, interprofessional health care team members. Communicating effectively also involves effective use of conflict management and negotiation. Provides effective 'hand off' or transitional communication using appropriate tools/strategies (iSBAR, iSHAPED).</i>
19	<i>Communicates therapeutically.</i>	<i>Establishes therapeutic presence and rapport with client. Actively listens to client. Demonstrates a caring presence and engages therapeutically in both verbal and nonverbal communication. Uses a variety of communication strategies as necessary.</i>
20	<i>Documents clearly, concisely, and accurately</i>	<i>Accurately and concisely reflects care in the electronic health record.</i>
Category: Quality and Safety		
21	<i>Integrates informatics to support decision-making</i>	<i>Uses resources available (policies and procedures, point of care references) to inform decision making. Additionally, utilizes technology to identify pertinent information (electronic health record, online references and point-of-care resources from credible sources, policy/procedures, etc.). According to the American Nursing Association, "Nursing informatics (NI) is the specialty that integrates nursing science with multiple information management and analytical sciences to identify, define, manage, and communicate data, information, knowledge, and wisdom in nursing practice."</i>

		<i>Nursing informatics incorporates the fields of nursing, computer science, and information science in order to manage medical data and develop and maintain data systems that are designed to improve patient outcomes as well as boost the overall performance of a healthcare organization</i>	
22	<i>Adheres to principles of patient safety</i>	<i>Identifies patient using patient identifiers, uses sterile technique, washes hands, adheres to isolation or fall precautions as appropriate.</i>	<i>Identifies patient using patient identifiers, uses sterile technique, washes hands, adheres to isolation or fall precautions as appropriate. Ordering appropriate prescriptions considering various aspects that could impact patient.</i>
23	<i>Advocates for client</i>	<i>Engages in opportunities to advocate for client to alternate audiences (providers, nurses, family members, agency personnel, policymakers, leadership, etc.) as appropriate to situation.</i>	
Category: Professionalism			
24	<i>Adheres to legal and ethical standards.</i>	<i>Acts in accordance with the American Nurses Association Code of Ethics for Nurses, adheres to legal regulations and standards, functions at maximum level of leadership within scope of practice, recognizes ethical implications of decisions, utilizes ethical decision-making frameworks.</i>	
25	<i>Demonstrates professionalism.</i>	<i>Acts in accordance with the definition provided above, able to accept and incorporate instructor feedback, comes to clinical experience well prepared, not impaired (sleep deprived and/or other impairment), dressed professionally, conducts self in professional manner. Demonstrates leadership behaviors in clinical situations, navigates conflict professionally, maintains commitment to life-long learning.</i>	

These 25 items were meticulously identified through content validity surveys and expert focus groups.

(Could link to our Conceptual Framework and Content Validity manuscript here).

Transition to new topic..... ? Some kind of visual here?

Now that you have a solid understanding of the structural framework and the operational definitions, let's move into a discussion of the **scoring of the CCEI 2.0 which is grounded in a dichotomous model for accuracy and consistency.** This model uses clear, binary

decision points: 0 for below threshold of competency and 1 for meets threshold of competency, with N/A for not applicable. Faculty determine performance indicators prior to evaluations to ensure consistent, objective assessments. Performance indicators, formerly called expected minimum behaviors, represent observable actions that students must demonstrate in order to achieve the 1 associated with each item of the CCEI 2.0.

(could show visual here for content below – that shows examples of what majority scoring, majority scoring including a critical element, and all or none scoring).

There are **two approaches to using this scoring system** and schools can determine which option is most appropriate for your school and your philosophy on the assessment of student performance. The first approach is the '**majority**' approach, where students must achieve a majority of the identified performance indicators to earn the point. For example, if there are five indicators for 'Recognize Cues,' faculty would determine the minimum number needed to score that item. A component of this scoring method could include the identification of '**critical elements**,' items that must be demonstrated in order to earn the point. The second is the '**all or none**' approach, where students must achieve all identified performance indicators to earn the point. **Both are acceptable, but faculty must reach consensus on the performance indicators and the scoring method to ensure consistency among faculty graders prior to the assessment activity.**

Sub scores for categories and subcategories are calculated by dividing points earned by points possible, and the total competency score is also calculated this way.

Institutions set their own passing scores based on their grading scale. This approach minimizes bias and aligns with AACN and NLN guidelines for clear, measurable behaviors in competency-based education.

(Visual) could show a CCEI2.0 scoring sheet filled out at various levels – and move through those visuals. - filled out at the item level (and maybe with a comment written on the side) - then filled out with some items and enough to complete a subcategory or two. - Then filled out through a couple of categories – and then the completed CCEI2.0 filled out in its entirety.

Formative vs summative assessment

Assessment in nursing education plays a critical role in evaluating student learning and ensuring competency in clinical practice. **Formative assessment** is an ongoing process that provides real-time feedback to students, allowing them to improve their knowledge, skills, and decision-making abilities throughout their learning experience. The CCEI 2.0

offers a structured framework for formative assessment, fostering active learning through immediate feedback in a low-stakes environment. Faculty can provide consistent feedback using standardized terminology, and by integrating language from the Clinical Judgement Measurement Model, they can reflect on students' clinical judgment process. This approach allows educators to tailor future teaching/learning activities and to more accurately pinpoint areas in need of remediation ensuring students develop the clinical judgment skills necessary for safe and effective patient care.

Summative assessment, on the other hand, occurs at the conclusion of a learning period and measures a student's achievement of specific competencies or learning outcomes. Summative assessments **determine whether students have met the required level of proficiency** before advancing to higher levels of practice or entering professional roles. **By aligning scenarios with students' educational levels and adjusting performance indicators on the CCEI 2.0 rubric to reflect expected competencies at each stage, faculty can effectively assess student growth across the curriculum at various time points.** This structured approach ensures that competency development is measured progressively, supporting students' readiness to practice.

In **competency-based education (CBE)**, both formative and summative assessments are essential in guiding students toward mastery of clinical skills and decision-making. By integrating both assessment types, the **CCEI 2.0 supports a comprehensive evaluation approach that aligns with the principles of CBE, ultimately ensuring that nursing students are prepared for real-world clinical responsibilities.**

Conclusion

CCEI 2.0 is an excellent tool for providing structured feedback to students. It incorporates terminology essential to competency-based education and ensures clear, consistent expectations through its performance indicators. This clarity helps students understand that assessments are consistent across their peers and throughout their program.

In conclusion, this module has outlined the structure and scoring of the CCEI 2.0. As we move through the modules, we'll delve deeper into achieving faculty consensus on performance indicators and scoring details.

Thank you for joining me, and I look forward to our next session together!